

Comparing Deep Learning Models for Hotel Demand Forecasting and Dynamic Room Pricing

Jack DeMarinis
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United States

Abstract—This project develops an end-to-end hotel demand forecasting and pricing-support pipeline using real operational hotel data. Raw reservation, room-sales, and forecast reports are cleaned, deduplicated, audited, and converted into a daily modeling table covering 2023–2025. The main forecasting target is daily rooms sold. Classical baselines, tree-based tabular models, and neural sequence models are evaluated using chronological train, validation, and test splits. The primary reported model convention excludes operational rows flagged as known anomalies. XGBoost is the best model by test mean absolute error (MAE), achieving MAE 7.443, RMSE 9.927, and R^2 of 0.585 on the unflagged 2025 test period. Deep learning models are competitive but do not surpass the best tree models: the best GRU reaches MAE 7.813, and the best Transformer reaches MAE 7.835. A bounded pricing layer converts the selected XGBoost demand forecast into aggressive and conservative ADR recommendations. Under a static counterfactual that holds realized rooms sold fixed, the aggressive scenario increases simulated unflagged 2025 room revenue by \$88,101.38 and RevPAR by \$4.15, while the conservative scenario increases simulated revenue by \$44,092.35 and RevPAR by \$2.08. The results support a practical MVP conclusion: tree-based tabular forecasting remains the strongest approach for this small daily dataset, while GRU and Transformer models provide useful deep learning comparisons.

Index Terms—hotel demand forecasting, revenue management, XGBoost, random forest, GRU, LSTM, Transformer, dynamic pricing, interpretability

I. Introduction

Hotels make pricing decisions under uncertainty: demand varies by day of week, season, events, booking behavior, room availability, and prior occupancy. Many small or independent hotel operations still rely on spreadsheets, manual review, or fixed pricing rules, even though commercial revenue management systems use data-driven demand signals to update prices more systematically. The goal of this project is to build an academic minimum viable product (MVP) that forecasts daily hotel demand and converts that forecast into bounded pricing recommendations.

The project is motivated by a real business setting using historical operational hotel reports. From a machine learning perspective, the task is a structured time-series regression problem. From a business perspective, it is a pricing-support problem where prediction errors and recommendation constraints both matter. The project therefore evaluates predictive performance using MAE, RMSE, MAPE, and R^2 , and evaluates pricing support

using static revenue, ADR, occupancy, and RevPAR metrics.

The proposal emphasized comparison between classical machine learning and deep learning architectures. This report implements that comparison using simple time-series baselines, Ridge regression, Random Forests [1], XGBoost [2], MLP models, LSTM models [3], GRU models [4], and small Transformer encoders [5]. The final system uses a two-stage pipeline: first forecast demand, then apply bounded pricing rules.

II. Data and Business Setting

The data come from D. Hotel Suites & Spa, an independent 60-room, six-floor hotel located at 1 Country Club Road in Holyoke, Massachusetts. The property serves regional travel demand in the Pioneer Valley, sitting near Route 5 and Interstate 91 between Northampton, Amherst, and Springfield, and roughly forty minutes from Bradley International Airport. Inventory is consistently 60 rooms across all reporting periods, which is the property capacity used throughout the modeling pipeline. The hotel currently sets nightly rates manually, using spreadsheets, historical reference rates, and operator judgment, with no automated demand forecast and no automated pricing recommendation. This is a common situation for independent hotels of this size, because commercial revenue management systems are designed for larger chains and tend to be expensive to integrate for a single small property. The MVP described in this paper is being developed for D. Hotel Suites & Spa as an internal, lower-cost alternative to a commercial system. The author has access to the property’s operational reports through a family connection (a partner at the hotel), which is the basis for using a real single-property dataset rather than a public benchmark; further detail on the property and the business motivation is given in Appendix A.

The raw data consist of operational workbooks exported from the property’s reservations and reporting system, including reservation-style guest reports, Block RoomSales reports, and ReservationForecast reports.

The final modeling table contains 1,096 daily rows and 209 columns covering January 1, 2023 through December 31, 2025. The table is date-grain: each row represents one business date. It joins cleaned room-sales history,

reservation forecast fields, and guest-arrival enrichment. The primary forecasting target is daily rooms sold.

Cleaning included parsing multi-line guest report blocks, standardizing dates and numeric fields, normalizing key text values, preserving source lineage, and removing duplicates with an auditable deduplication process. The data quality audit flagged two over-capacity rows and twelve zero-room-sold rows. These rows are retained for traceability, but the primary reported model convention excludes all flagged dates because they are known operational anomalies.

Fig. 1 summarizes the final pipeline.

III. Feature Engineering

Features were designed to be forecast-oriented and leakage-safe. Same-day operational fields, same-day guest enrichment fields, and raw future ReservationForecast fields were excluded from learned model inputs. The model inputs combine:

- calendar features such as day of week, month, quarter, weekend flags, and cyclical encodings;
- lagged historical values for rooms sold, ADR, revenue, and occupancy;
- rolling means, medians, and standard deviations over 7, 14, 28, and 56 day windows;
- derived comparisons such as week-over-week changes.

All rolling features are shifted so that the current day’s target is not included in the feature values. This matters because the intended use case is forecasting before final realized demand is known.

IV. Models and Evaluation

A. Splitting Strategy

The dataset is split chronologically. Training uses January 1, 2023 through June 30, 2024; validation uses July 1, 2024 through December 31, 2024; and testing uses calendar year 2025. This matches the operational goal of generalizing to a later time period.

Three data-quality conventions are reported. The full-data reference uses all rows. The over-capacity sensitivity removes only rows where rooms sold exceeded inferred capacity. The primary anomaly-excluded convention drops every row tagged by the data-quality audit, and is the convention used for the headline results throughout the paper.

B. Classical and Tree-Based Models

Baselines include previous-day demand, previous-week demand, rolling averages, and Ridge regression. Tree-based tabular models include Random Forest and XGBoost. The implementation uses scikit-learn [6] and XGBoost.

Table I shows the full-data 2025 test results. XGBoost is best by MAE, while Random Forest has slightly stronger RMSE and R^2 . MAPE values should be read with care: the property has only 60 rooms and the test period

TABLE I
Full-Data 2025 Test Metrics

Model	Rows	MAE	RMSE	MAPE (%)	R^2
XGBoost	365	8.039	11.210	45.42	0.539
Random Forest	365	8.062	10.870	42.36	0.567
Previous Week	365	9.551	13.438	35.02	0.338
Previous Day	365	11.307	15.987	42.01	0.063
Rolling 7d	365	12.528	15.107	44.45	0.164
Ridge	365	12.744	14.938	52.08	0.182
Rolling 28d	365	13.511	16.073	52.46	0.053

TABLE II
Best Learned Model by Data-Quality Convention

Scenario	Model	Rows	MAE	RMSE	MAPE (%)	R^2
All flags excl.	XGBoost	354	7.443	9.927	45.62	0.585
Over-cap excl.	RF	365	8.019	10.757	41.61	0.576
Full data	XGBoost	365	8.039	11.210	45.42	0.539

contains several low-demand and near-zero-demand dates, so MAPE is sensitive to small denominators and tends to look pessimistic relative to absolute-error metrics. MAE and RMSE are therefore treated as the primary comparison metrics in this report, with MAPE reported for completeness.

Table II shows the sensitivity analysis. The primary report result is XGBoost under the anomaly-excluded convention: MAE 7.443, RMSE 9.927, and R^2 0.585.

Fig. 2 compares predicted and actual daily rooms sold across the 2025 test period for XGBoost, Random Forest, and the two strongest heuristic baselines. The learned tabular models track the weekly seasonal cycle and high-occupancy weekends closely; the heuristic baselines either lag the signal (previous-day) or smooth out short-term swings (previous-week).

C. Deep Learning Models

The deep learning comparison includes tabular MLP models and sequence models. LSTM, GRU, and Transformer candidates use sequential windows ending on each prediction date. PyTorch [7] is used for LSTM, GRU, and Transformer training, with AdamW optimization [8], Smooth L1 loss, gradient clipping, and early stopping on validation loss.

1) Hyperparameter Sweep: Fourteen configurations were trained across the four families: three MLP variants, three LSTM variants, six GRU variants, and two Transformer variants. The search axes are summarized in Table III. Lookback length was the primary search axis for the sequence models. All sequence configurations use one recurrent (or Transformer) layer; Transformer variants use four attention heads. Dropout is zero for LSTM and GRU and 0.10–0.15 for Transformer. Patience for early stopping ranges from 90 to 120 epochs depending on the configuration. Targets and feature inputs are standardized before training. The sweep is intentionally scoped: the goal

Hotel Demand Forecasting & Pricing MVP — Pipeline Architecture

Three-stage flow: operational data → demand forecasting → bounded pricing recommendations

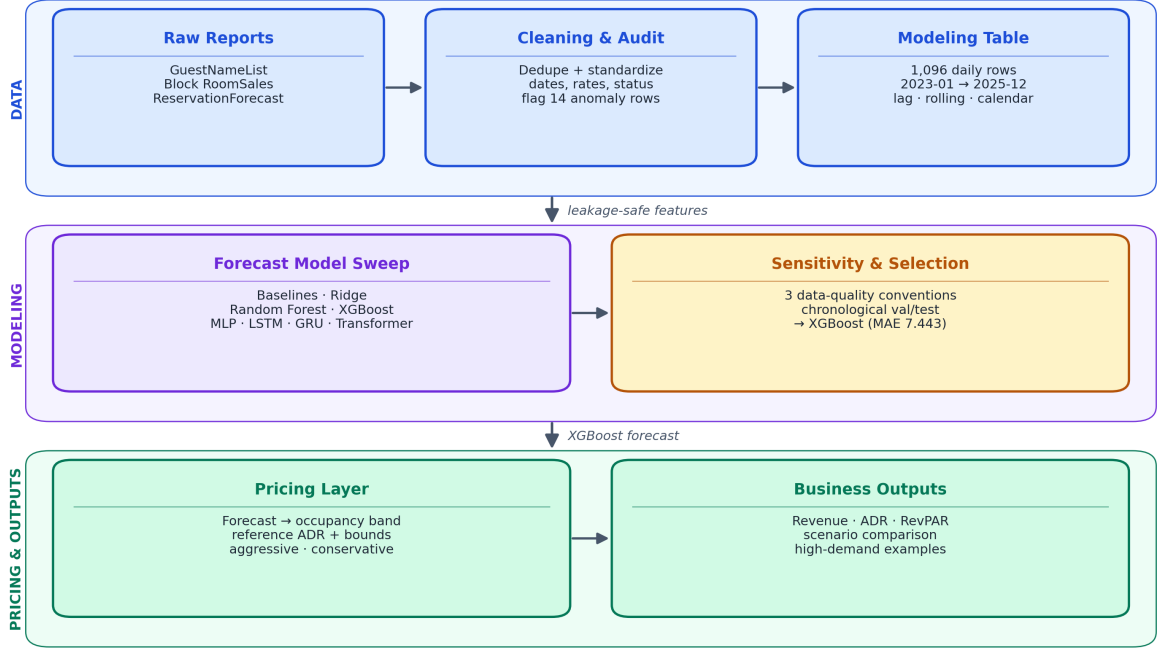


Fig. 1. End-to-end hotel demand forecasting and pricing MVP pipeline.

TABLE III
Deep Learning Hyperparameter Sweep

Family	Lookback (days)	Hidden size	Learning rate	Configs
MLP	–	(32), (64,32)	0.0007–0.001	3
LSTM	14, 28, 56	32, 48, 64	0.0005–0.001	3
GRU	14, 28, 42, 56, 84	32, 48, 64	0.0005–0.001	6
Transformer	28, 56	48, 64 (4 heads)	0.0004–0.0005	2

TABLE IV
Best Model per Family on Unflagged 2025 Test Rows

Model	Rows	MAE	RMSE	MAPE (%)	R^2
XGBoost	354	7.443	9.927	45.62	0.585
Random Forest	354	7.466	9.834	42.67	0.593
GRU 56d wide	354	7.813	10.588	48.82	0.528
Transformer 28d	354	7.835	10.305	43.23	0.553
LSTM 56d	354	8.662	11.210	51.68	0.471
MLP medium	354	12.663	16.631	64.36	−0.165

is an honest comparison rather than an exhaustive neural architecture search.

2) **Headline Results:** Table IV compares the strongest model from each architecture family under the primary anomaly-excluded test convention, including all four deep learning architectures from the proposal. The best GRU reaches MAE 7.813, and the best Transformer reaches MAE 7.835. The best LSTM reaches MAE 8.662, and the best MLP only reaches MAE 12.663 with a negative R^2 , indicating that purely tabular MLPs do not generalize on this small dataset.

3) **Statistical Significance:** The point-estimate MAE differences between XGBoost and the strongest sequence models are small relative to the size of the test set, so the headline ranking deserves a statistical check. Table V reports paired bootstrap 95% confidence intervals on the test MAE difference (other model minus XGBoost) using

10,000 resamples of the 354-row unflagged 2025 test set, drawn with replacement at the date level. The Random Forest, best GRU, and best Transformer intervals all span zero, so under this paired date-level bootstrap the XGBoost advantage over those models is not statistically distinguishable on this test set. Only the best LSTM and the best MLP intervals exclude zero, meaning XGBoost is significantly better than those families at 95% confidence under the same procedure but only nominally better than Random Forest, GRU, and Transformer. This is the more accurate way to read Table IV: under this paired date-level bootstrap the strong tree and best sequence models form a statistical tie, while LSTM and MLP variants tested here trail meaningfully.

Fig. 3 visualizes the model comparison. The honest

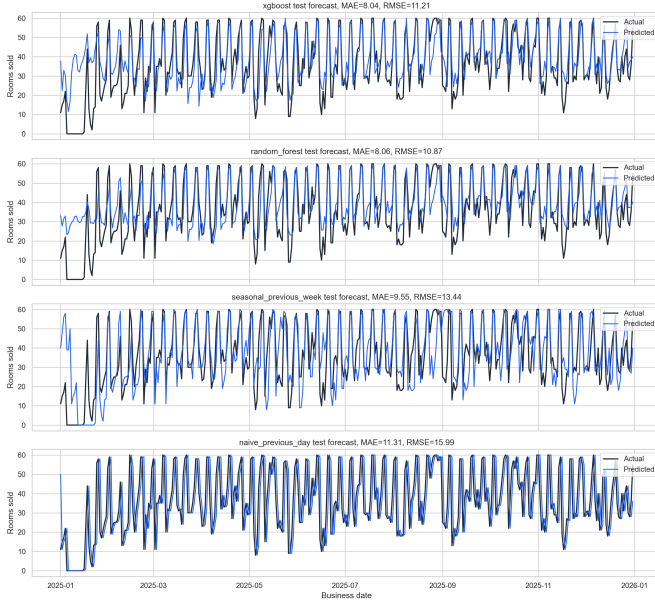


Fig. 2. Predicted versus actual daily rooms sold over the full 2025 test period for the strongest tabular models and two reference baselines.

TABLE V
Paired Bootstrap 95% CIs vs. XGBoost (10,000 Resamples, 354 Rows)

Comparison	ΔMAE	95% CI
Random Forest	+0.023	$[-0.315, +0.349]$
Best GRU	+0.369	$[-0.203, +0.925]$
Best Transformer	+0.392	$[-0.215, +0.978]$
Best LSTM	+1.219	$[+0.656, +1.789]$
Best MLP	+5.220	$[+4.090, +6.324]$

interpretation is that sequence neural networks are statistically competitive with tree-based tabular models on this dataset, but the available sample size does not give a sequence architecture a significant advantage over XGBoost. The deep learning hypothesis from the proposal is therefore supported in part: GRU and Transformer match the strongest tabular baselines, but they do not surpass them.

V. Feature Importance

The XGBoost feature-importance analysis shows that recent weekly history is the strongest signal. The top features are the seven-day and fourteen-day lags of rooms sold, the seven-day lag of net room revenue, the day-of-week encoding, and the one-day lag of net room revenue. This is consistent with the domain expectation that hotel demand has strong weekly seasonality and short-term persistence.

VI. Interpretability and Model Choice

Under the paired date-level bootstrap reported in Table V, XGBoost, Random Forest, the best GRU, and the best Transformer are statistically indistinguishable on

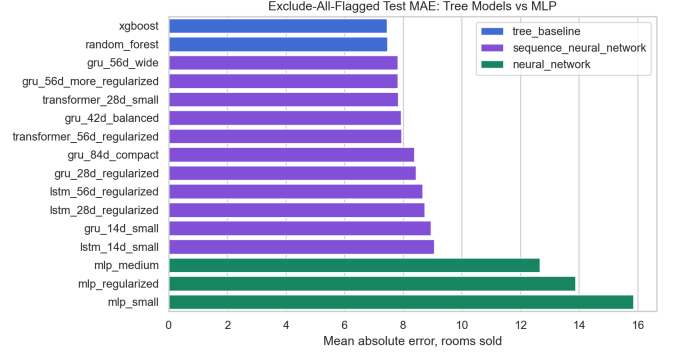


Fig. 3. Exclude-all-flagged test MAE comparison across tree and neural models.

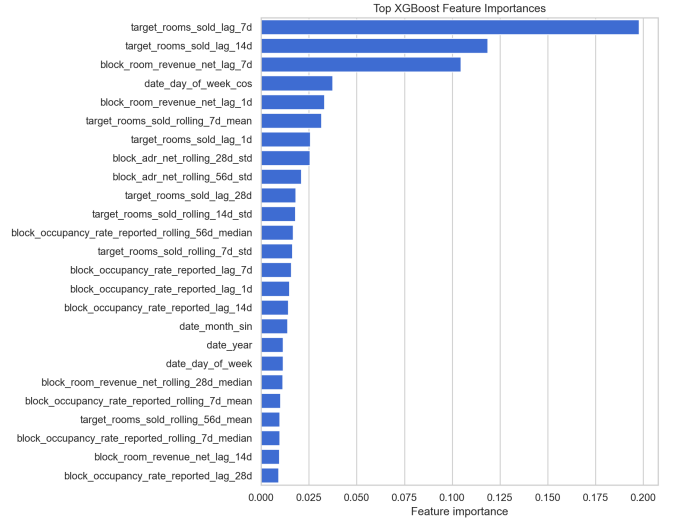


Fig. 4. Top XGBoost feature importances under the primary training setup.

test MAE. When predictive accuracy is indistinguishable, principled model selection requires a second axis. For a pricing-support MVP that surfaces ADR recommendations to a human revenue manager, that second axis is interpretability: a recommendation is operationally useful only if the manager can audit why the model is asking to raise or lower the rate, and an unexplained ADR change is hard to distinguish from a guess. This framing follows the position taken in high-stakes interpretable-ML work [9], which argues that models affecting real-world decisions should be inherently or post-hoc explainable rather than treated as opaque oracles.

The model families differ substantially on this axis. The heuristic baselines are trivially interpretable but are too inaccurate for production use. Ridge regression provides global linear coefficients but underperforms by a wide margin and cannot capture the nonlinear interactions present in hotel demand. Random Forest and XGBoost are tree ensembles whose individual trees are inspectable and

which support mature post-hoc explanation tools: feature importance, partial dependence, and local Shapley-value attributions [10] that map any single prediction back to the features that drove it. The MLP, LSTM, and GRU sequence models are less directly interpretable: post-hoc methods such as permutation importance, LIME [11], integrated gradients, and SHAP applied to the neural model are possible but require additional engineering, and their explanations are noisier on a 1,096-row dataset. Transformer attention weights provide a partial view of which past timesteps the model attends to, but attention is not a direct substitute for feature attribution.

Combining the bootstrap and interpretability views resolves the model choice more cleanly than the raw MAE table alone suggests. XGBoost is statistically tied with the strongest sequence models on accuracy but is materially more transparent. The feature-importance result in Fig. 4 already shows that the model’s strongest signals are seven-day and fourteen-day lagged rooms sold, lagged room revenue, and day-of-week encoding — features a hotel manager would recognize as the same signals they themselves use, which is a useful sanity check on the model’s logic. A SHAP-based per-prediction explanation could be added without retraining, because tree-specific SHAP implementations exist for XGBoost. The same level of explanation would require additional engineering for the GRU or Transformer with no accuracy gain in return. Under the criterion that the model must support human-approved pricing decisions, the tie among XGBoost, GRU, and Transformer on accuracy resolves in favor of XGBoost on interpretability. The recommended choice is therefore not “XGBoost wins by MAE” but “XGBoost is the right operational choice because it is statistically as accurate as the best sequence models while being more auditable for a revenue manager.”

VII. Pricing Recommendation Layer

The pricing layer uses the selected XGBoost forecast. Forecast occupancy is computed as forecast rooms sold divided by total property rooms. A reference ADR is chosen from historical same-day ADR with rolling and lagged ADR fallbacks. The pricing rule then applies bounded demand-based adjustments.

Two scenarios are evaluated. The aggressive scenario uses 4%, 8%, and 12% adjustment bands. The conservative scenario uses 2%, 4%, and 6% bands. ADR is also bounded by minimum and maximum ADR constraints and by a maximum daily change from the reference ADR.

The pricing simulation is intentionally static. It applies recommended ADR to actual realized rooms sold and does not estimate price elasticity, competitor reaction, or occupancy changes caused by price changes. This makes the result a useful academic counterfactual, not proof of actual achievable revenue.

Table VI shows static unflagged 2025 pricing results. The aggressive scenario produces a simulated revenue lift

TABLE VI
Static Pricing Scenario Results on Unflagged 2025 Test Days

Scenario	Hist. Rev.	Sim. Rev.	Delta	Delta %
Aggressive	\$2.018M	\$2.106M	\$88.1K	4.37%
Conservative	\$2.018M	\$2.062M	\$44.1K	2.18%

TABLE VII
Business Metrics on Unflagged 2025 Test Days

Scenario	Act. Occ.	Fcst. Occ.	Hist. RevPAR	Sim. RevPAR
Aggressive	63.63%	66.13%	\$95.01	\$99.16
Conservative	63.63%	66.13%	\$95.01	\$97.09

of \$88,101.38, while the conservative scenario produces \$44,092.35.

Table VII adds occupancy and RevPAR context. Static occupancy delta is zero by design because rooms sold are held fixed. RevPAR improves by \$4.15 in the aggressive scenario and \$2.08 in the conservative scenario.

Fig. 5 compares historical and recommended ADR over time. Fig. 6 compares the aggressive and conservative revenue deltas.

VIII. High-Demand Scenario Examples

To make the pricing recommendations concrete, Table VIII shows high-demand weekend dates from the unflagged 2025 test period. These dates have forecast occupancy near sellout levels and therefore receive upward ADR recommendations. For example, October 25, 2025 had 59 actual rooms sold and a forecast of 60.0 rooms; the aggressive rule recommends increasing ADR from \$218.36 to \$244.56, which yields a static simulated revenue delta of \$1,545.61.

IX. Limitations

There are several limitations. First, the pricing simulation does not model demand response to price changes. A real revenue management system would need elasticity estimates, competitor pricing, channel mix, and customer response modeling. Second, some proposal features such as competitor prices and event indicators were only included if available; the final dataset did not contain a reliable external competitor or event feed. Third, booking lead time and booking pace were proposed as candidate features but are not engineered in the final model, because the only multi-record reservation source covers 2025 only and using it as training history would overlap with the held-out test period. Fourth, the proposal allowed either date-grain or date-room-type-grain modeling; the final implementation uses date-grain only because the available daily sample size already limits deep learning generalization, and a per-room-type breakdown would further fragment the signal. Fifth, January 2025 zero-room-sold dates were flagged as likely operational anomalies and should not be used as primary business evidence without verification. Finally,

TABLE VIII
High-Demand Weekend Pricing Examples

Date	Day	Actual	Forecast	Hist. ADR	Agg. ADR	Cons. ADR	Agg. Delta
2025-10-25	Saturday	59	60.0	\$218.36	\$244.56	\$231.46	\$1,545.61
2025-08-09	Saturday	57	60.0	\$173.94	\$194.81	\$184.38	\$1,189.38
2025-11-15	Saturday	59	60.0	\$166.05	\$185.98	\$176.01	\$1,176.15
2025-12-06	Saturday	60	60.0	\$160.63	\$179.91	\$170.27	\$1,156.63
2025-11-22	Saturday	58	60.0	\$161.24	\$180.59	\$170.91	\$1,122.11
2025-05-24	Saturday	57	59.9	\$237.02	\$265.46	\$251.24	\$1,621.31

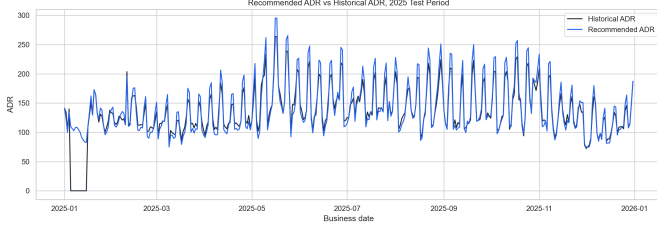


Fig. 5. Recommended ADR versus historical ADR during the 2025 test period.

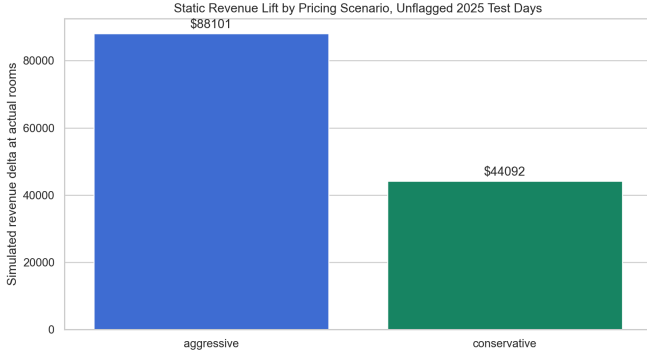


Fig. 6. Static simulated revenue lift by pricing scenario.

the dataset is small for deep learning: the best GRU and Transformer models get close, but the tabular tree models remain better suited to the available sample size.

X. Conclusion

This project implements the proposed hotel demand forecasting and pricing-support MVP end to end. The final system cleans operational reports, builds a daily modeling table, audits data quality, compares classical baselines against deep learning models, selects a primary demand model, and converts forecasts into bounded pricing recommendations.

The strongest forecasting result is XGBoost under the anomaly-excluded convention, with test MAE 7.443, RMSE 9.927, and R^2 0.585. Deep learning models provide a meaningful comparison: the best GRU reaches MAE 7.813 and the best Transformer reaches MAE 7.835. Under a paired date-level bootstrap on the 354-row test set, the XGBoost advantage over Random Forest, the best GRU,

and the best Transformer is not statistically significant at 95% confidence, while the advantage over the best LSTM and best MLP is. The honest summary is therefore that the strongest tabular and sequence models tested form a statistical tie on this dataset, with LSTM and MLP variants trailing significantly. The pricing layer shows practical business potential under static assumptions, with simulated unflagged 2025 revenue lifts of 4.37% for the aggressive scenario and 2.18% for the conservative scenario.

Overall, the final recommendation is to use XGBoost as the MVP forecasting model. The justification is no longer purely accuracy-based: among the strongest models, accuracy ties at 95% confidence under this paired date-level bootstrap, and XGBoost is selected because it is the most interpretable of the tied models and therefore the easiest for a revenue manager to audit and approve. GRU and Transformer results are presented as strong deep learning comparisons that match XGBoost’s accuracy but require additional explanation tooling for operational use. The pricing layer is treated as a bounded decision-support tool rather than a complete commercial revenue management system.

Appendix A Property and Business Context

The MVP target property is D. Hotel Suites & Spa, an independent 60-room hotel in Holyoke, Massachusetts. The building has six floors and is located at 1 Country Club Road, near Route 5 and Interstate 91, on the edge of Holyoke Country Club. The property serves Pioneer Valley travel demand, with the most relevant nearby drivers being Northampton (immediately adjacent), Amherst (about fifteen minutes north), Springfield (about fifteen minutes south), and Bradley International Airport (about forty minutes south). The hotel was previously operated under the Country Inn & Suites Holyoke flag and has since been rebranded as an independent property. Inventory consists of a regular room class and a suite class, totaling 60 rooms; on-property amenities include a spa, indoor pool, fitness center, breakfast service, and a bar and restaurant. The 60-room inventory figure is consistent across every operational reporting period in the dataset and is used directly as the property capacity throughout the modeling and pricing pipeline. The two over-capacity

rows flagged in the data-quality audit correspond to dates where the source report shows rooms sold above 60, which is mechanically impossible given the inventory and is therefore treated as an operational reporting anomaly rather than real demand.

The business motivation for this project is direct and specific. The property currently sets nightly room rates entirely by hand, using spreadsheets, historical patterns, and operator judgment. There is no automated demand forecast and no automated pricing recommendation; every adjustment is a manual decision made by the team that runs the property. This is a common situation for independent hotels of this size, because commercial revenue management systems are generally designed for chains or larger inventories and tend to carry setup, integration, and licensing costs that are difficult to justify for a single 60-room property. The MVP described in this paper is intended to fill that gap: it produces internally usable demand forecasts and bounded ADR recommendations from the same operational reports the team already runs, and it deliberately keeps the recommendation step bounded and human-approved rather than fully automated. The author’s family connection to a partner at the property is what made it possible to use this particular dataset for an academic project; the data are not publicly available, but the modeling code, evaluation harness, and report-generation pipeline are open at the repository linked in Appendix B, with personally identifiable guest information excluded from the modeling artifacts. The research goal of the proposal — comparing classical machine learning and deep learning architectures on hotel demand forecasting — and the practical goal of the property — replacing fully manual pricing with a model-supported, manager-approved recommendation flow — are aligned, because both require an accurate forecast that an operator can audit and trust before applying it to live rates.

Appendix B

Repository, Reproduction, and Artifacts

The full source code, cleaning scripts, modeling pipeline, pricing layer, and report artifacts are publicly available at https://github.com/jackdemarinis/CSC_561_Final_Project. Raw workbooks live in `Data/`; cleaned outputs in `cleaned_data/`; derived modeling features in `processed_data/`; metrics, figures, and the IEEE report under `reports/`. Source code is split under `src/` into feature construction, model training, evaluation, and pricing. Tables and figures in the paper are generated from the same CSV and JSON artifacts produced by the scripts.

The reproduction order is: clean operational exports (`scripts/clean_new_operational_reports.py`); build guest daily enrichment and the date-grain modeling table (`src/features/`); audit data quality (`src/evaluation/audit_modeling_data_quality.py`); train baselines, run sensitivity, and train deep learning models (`src/models/`,

`src/evaluation/run_baseline_sensitivity.py`); build pricing recommendations (`src/pricing/`); and compile reports/final/final_report_ieee.tex. The central artifact is processed `data/modeling_daily.csv`: 1,096 daily rows from 2023-01-01 through 2025-12-31, with no missing or duplicate business dates. Train ends 2024-06-30; validation runs through 2024-12-31; test is calendar year 2025.

Result tables and figures map directly to subfolders: `reports/audit/` for data-quality flags, `reports/baselines/` and `reports/deep_learning/` for model metrics, `reports/pricing/` for ADR recommendations and business metrics, and `reports/figures/` for PNG figures. Inspecting those CSV/JSON files is the quickest way to check whether the PDF has drifted from current code output.

Appendix C

Data Sources and Modeling Policy

A. Data Sources

Three operational families feed the modeling table. Block RoomSales (2023–2025) provides daily property-level rooms sold, revenue, ADR, RevPAR, and inventory; reconciliation checks confirm room-type rows plus deduplicated adjustment rows tie to property daily totals, and inferred hotel inventory is 60 rooms throughout. ReservationForecast files (2023–2024) overlap exactly with Block on rooms sold, out-of-order rooms, and net room revenue, but use a different occupancy denominator (sellable rooms vs. total inventory); both definitions are kept as explicit columns rather than collapsed. GuestNameList exports are 2025-only reservation reports parsed into 10,579 deduplicated stay-like records. Guest PII (names, emails, confirmation/folio numbers) is used only for cleaning and deduplication and is excluded from modeling features; the daily enrichment aggregates to non-identifying counts, shares, and rates.

B. Feature Policy and Leakage Control

The forecasting task uses calendar features (date fields, cyclical day-of-week, month, and day-of-year encodings), lag features at 1, 7, 14, 28, and 365 days, and rolling windows over 7, 14, 28, and 56 days, all over rooms sold, net ADR, net room revenue, and reported occupancy. Same-day Block fields, raw ReservationForecast fields, same-day guest enrichment, and source lineage columns are blocked from learned model inputs unless they appear only as lagged or rolling historical versions. Guest-derived features are also treated cautiously because the guest-list source mainly covers 2025 and is not a complete multi-year historical signal. This leakage policy is the most important experimental safeguard in the repository.

C. Data-Quality Sensitivity

Fourteen anomaly rows are flagged in the modeling table: twelve zero-room-sold dates and two rows where rooms sold exceed the 60-room inventory. The primary

exclude_all_flagged convention drops all fourteen, leaving 354 test rows in 2025. The decision-level sensitivity is stable: XGBoost is best by MAE under both the exclude-all-flagged and full-data conventions, and Random Forest is best under the over-capacity-only convention. Random Forest is sometimes stronger by RMSE or R^2 , which is why both tree models remain visible in the main comparison.

Appendix D

Pricing Layer Notes

The pricing layer is a bounded decision-support rule, not a revenue management optimizer. It starts from the selected XGBoost forecast, converts forecast rooms sold into forecast occupancy against the 60-room inventory, assigns a demand band, and adjusts a reference ADR drawn from historical same-day net ADR with rolling, lagged, and median fallbacks. Recommendations are bounded by minimum ADR, maximum ADR, and a 20% daily move cap from the reference. Two scenarios (aggressive and conservative) are reported because a single pricing curve would imply false precision; both are static counterfactuals applied to actual realized rooms sold and do not model price elasticity, competitor reaction, or channel mix.

A production version would need at least four extensions: (i) elasticity estimates or controlled experiments so ADR changes affect expected rooms sold rather than being applied to fixed history; (ii) competitor-rate, local-event, and holiday-demand feeds; (iii) error monitoring by day of week, occupancy band, month, and high-demand period; and (iv) human approval for flagged dates, unusually large ADR moves, and near-sellout dates. The current system can identify days when historical signals suggest higher or lower occupancy pressure and translate that into bounded ADR advice; it cannot yet answer whether a specific ADR change maximizes profit after customer response, cancellations, competitor movement, and distribution costs.

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